

ELIJAH, OLUWAPELUMI ISRAEL

M79, General Hospital Road, Ipetu Ijesa, Osun State, Nigeria | +234 8146433876 | +234 8148473095

elijaholuwapelumi3@gmail.com | [linkedin.com/in/oluwapelumi-israel-elijah](https://www.linkedin.com/in/oluwapelumi-israel-elijah)

RESEARCH INTEREST

Driven to advance geospatial technologies to address environmental, climatic, and security challenges. With a strong foundation in Remote Sensing and GIS, my research interests span earth observation, spatial intelligence, and data-driven environmental analysis.

Interests: Geospatial Data Science & Earth Observation | Dense Time Series Remote Sensing & Environmental Change Detection | Flood Risk & Environmental Monitoring | Machine Learning Applications

EDUCATION

Federal University of Technology Akure (FUTA), Akure, Ondo State, Nigeria 2020 – 2025

Bachelor of Technology (B. Tech) in Remote Sensing & Geoscience Information System

Class of Degree: First - Class Honours

CGPA: 4.53/5.00 (90.6%)

Thesis Title: Geospatial Mapping and Analysis of Smuggling Routes in Saki West Border, Oyo State, Nigeria.

RESEARCH EXPERIENCE

- **Undergraduate Thesis:** Geospatial Mapping and Analysis of Smuggling Routes and Border Security Vulnerabilities in Saki West, Oyo State, Nigeria. 2025
- **Research focus:** Applied GIS and Remote Sensing techniques to analyze smuggling route dynamics and border security vulnerabilities in a high-risk cross-border region.
- **Methods:** Integrated Sentinel-2 imagery, SRTM DEM, and GRID3 socio-economic data using LULC classification, terrain modeling, cost surface and Least-Cost Path Analysis, AHP-based weighted overlay, and viewshed (visibility) analysis.
- **Key findings:** Revealed that gentle slopes, forested land cover, road connectivity, and proximity to markets significantly facilitate smuggling, while suitability modeling identified priority locations for surveillance towers and CCTV deployment.
- **Impact & future relevance:** Demonstrated the use of geospatial intelligence as a decision-support tool for surveillance infrastructure planning, project feasibility assessment, and optimized resource allocation in border security operations.

WORK EXPERIENCE

- **Remote Sensing & GIS Analyst – Angola Coastal Dynamics Mapping Project** 2025
 - Integrated GEBCO bathymetry and Landsat 9 imagery to assess coastal and seafloor conditions along Angola's coastline.
 - Produced bathymetric hillshade maps, 8 depth transects, composite profiles and 3D seafloor visualizations.
 - Applied NDWI to Landsat 9 imagery using Python in Google Colab to extract shorelines for spatial analysis.
- **Geospatial Intern – GIS & Mapping Regiment (June - November)** 2024
 - 49 Army Mapping and Surveying Command (Nigerian Army Engineers), Owode -Yewa, Ogun State, Nigeria.
 - Assisted in the production of topographic maps for military operations, planning, and training.
 - Contributed to the creation of boundary survey plans for land administration and defense infrastructure mapping.
 - Supported the production of institutional training maps used for operational simulation and field exercises.

- Geospatial Analyst Intern – POI Validation (Remote role)**
 Geospatial Research Limited, Lagos State, Nigeria 2024
 - Conducted Point of Interest (POI) validation across assigned locations (LGA-based) to ensure accurate spatial data representation.
 - Verified, updated, and corrected geospatial attributes to improve location-based services and mapping quality.
 - Ensured data integrity by performing field checks and geospatial QC tasks for large-scale digital maps.
- GIS Assistant – Road Mapping & Engineering Support (November – December)**
 Lagos Road Engineering Project, Lagos State, Nigeria 2024
 - Updated existing road networks and digitized newly constructed or missing roads using high-resolution imagery.
 - Created and corrected spatial features to improve road topology and network accuracy.
 - Populated and validated attribute tables for road classification, direction, and connectivity
- Academic Tutor – Geological Map Interpretation (AGY206)**
 Department of Mining Engineering, Federal University of Technology Akure (FUTA), Nigeria 2023
 - Organized and delivered tutorial sessions on Geological Map Interpretation for Mining students.
 - Prepared explanatory materials and supported students in understanding structural geology concepts and field-based map analysis.
- Subject Teacher (Full-Time)** 2022
 Excellent Primary & Secondary School, Ipetu-Ijesa, Osun State, Nigeria
 - Taught Basic Technology to junior secondary students and Agriculture/Geography to senior secondary students.
 - Developed lesson plans, assessed student performance, and helped improve subject Understanding across multiple levels.

SKILLS AND COMPETENCES

- Relevant Technical Skills:** Remote Sensing & Earth Observation, GIS Modeling & Spatial Analysis Land Use/Land Cover (LULC) Classification, Machine Learning (Spatial), Security & Intelligence Mapping
- Digital Skills** (proficiency levels): ArcGIS Pro (Advanced), QGIS (Advanced), Microsoft Excel (Advanced), Google Earth Engine (Intermediate), ERDAS Imagine (Intermediate), ENVI (Intermediate), SNAP (Intermediate), Python (Intermediate), SQL (Intermediate), Microsoft Power BI (Intermediate), ViewSpec Pro (Basic)
- Relevant Soft Skills:** Analytical Thinking & Problem Solving, Research & Report Writing, Communication & Presentation, Leadership & Team Collaboration, Project Management & Organization, Adaptability & Innovative Thinking
- Language:** English (Proficient in Reading, Writing, Listening & Speaking), Yoruba (Native Speaker)

CERTIFICATIONS AND PROFESSIONAL DEVELOPMENT & MEMBERSHIPS

- Developing Sustainable Earth Science Applications (NASA's – ARSET), 2026** [View here](#)
- Fundamentals of Remote Sensing (NASA's – ARSET), 2026** [View here](#)
- Certificate of Membership, ISPRS Student Consortium 2026 (ISPRS SC)** [View here](#)
- Going Places with Spatial Analysis - ESRI MOOC, 2025** [View here](#)

- **Data Analytics Training Program** - The Data Immersed (TDI), 2025 [View here](#)
- **Climate-Smart Forestry** - FAO e-Learning Academy, 2024 [View here](#)

CONFERENCE/WORKSHOP ATTENDED

- **Open-Source GeoAI for Earth Observation Webinar - (ESGRS)** [LINK](#) 2026
 - Open-Source GeoAI for Earth Observation: Practical Tools for AI-Driven Analysis and Visualization.
- **Artificial Intelligence and Machine Learning for Agriculture Analytics - (ESGRS)** [LINK](#) 2026
 - Applied AI/ML techniques for geospatial crop monitoring and mapping using optical and SAR satellite imagery, with applications in precision agriculture and crop classification.
- **GIS Day Conference (FUTA Edition)** - Federal University of Technology Akure 2025
 - Mastering Spatial Logic: Developing Advanced Spatial Reasoning Skills for Proactive, Sustainable Planning.
- **GIS Day Conference (FUTA Edition)** - Federal University of Technology Akure 2024
 - Integration of Drone Technology for Precision Agriculture.
- **War Against Drug Abuse (WADA) Sensitization Workshop** - NDLEA 2024
Owode-Yewa Chapter, held at 49 Army Mapping & Survey Command (GIS Regiment)
 - Participated in a security-focused sensitization workshop addressing drug risk awareness for soldiers and student interns.
- **GIS Skill-Building Workshop** – NAPE-AAPG FUTA Chapter 2023
 - Unlock Your Potential with ArcMap and QGIS introductory and intermediate training on GIS tools for mapping and analysis

AWARDS, HONOURS & RECOGNITIONS

- **Geolevate Scholarship Award** 2025
Alumni Association of Remote Sensing & GIS Students
Awarded for academic excellence; received ₦100,000 support grant.
- **Award of Charitability** 2025
Head of Department, Remote Sensing & GIS
Presented in recognition of consistent dedication, outstanding performance, and positive impact within the department
- **Dean's List of Academic Excellence (6-time Honoree)** 2020–2025
Department of Remote Sensing & GIS Honored across multiple semesters for consistently ranking among top-performing students.
- **Sportsman of the Year** 2023
Department of Remote Sensing & GIS Department
Recognized for exceptional contribution and excellence in departmental sports activities.

- **Footprints Annual Scholarship Recipient** 2022
Footprints Foundation
Selected for outstanding academic performance and potential.

LEADERSHIP AND VOLUNTEERING EXPERIENCE

- **Special Adviser to the President (Media & Political Affairs)** 2023 – 2024
National Association of Ipetu - Ijesa Students (NAIIS)
 - Advised the NAIIS president on media communication strategies and student engagement.
 - Supported event planning, public relations, and policy discussions within the association.
- **Team Lead, Field Data Collection Trip – Akure South LGA** 2023 - 2024
 - Led a team of students during field data collection activities.
 - Coordinated GPS data gathering, field mapping tasks, and logistics.
 - Ensured accuracy of collected data and supervised team documentation.
- **Sports Director, Department of Remote Sensing & Geoscience Information System** 2022 - 2023
 - Coordinated departmental sports activities and competitions.
 - Organized inter-departmental events and ensured smooth logistics.
 - Led a team of student volunteers to promote student engagement through sports.
- **Volunteer Tutor – Mathematics & General Studies** 2020
Community-Based Teaching (COVID-19 Lockdown Initiative), Ipetu – Ijesa, Osun State, Nigeria.
 - Taught Mathematics and General Studies to junior secondary school during nationwide school closures.
 - Contributed to community learning continuity and supported the SDG goal on Quality Education by providing free academic guidance.

REFEREES

- REFERENCES AVAILABLE ON REQUEST